

NAME

extract_vdl2 – extract aircraft records from a dumpvdl2 log file by registration (REG) or ICAO hex address

SYNOPSIS

python3 extract_vdl2.py date4 REG_or_ICAO

DESCRIPTION

extract_vdl2 reads a dumpvdl2 VHF Data Link Mode 2 (VDL2) log file, locates every message block that belongs to a specified aircraft, and writes those blocks verbatim to a plain-text output file.

The aircraft to search for is identified either by its ICAO 24-bit hex address (e.g. 407739) or by its registration mark (e.g. G-TUMH). The tool auto-detects which type of identifier has been supplied: a string of exactly six hexadecimal characters is treated as an ICAO address; anything else is treated as a registration.

The input log file is located automatically by searching the current working directory for a file whose name contains the four-digit date suffix supplied on the command line (e.g. 0323 matches pi21vdl2_20260323.log).

ARGUMENTS

date4

Four-digit date string (MMDD) used to locate the log file. The tool searches the current directory for any *.log file whose name contains this string. Example: 0323.

REG_or_ICAO

Aircraft identifier. Supply either the ICAO 24-bit hex address (exactly six hexadecimal characters, e.g. 407739) or the registration mark (e.g. G-TUMH, EI-IKR). Case-insensitive.

LOG FILE FORMAT

The tool expects the standard output format produced by **dumpvdl2**. Each message block begins with a timestamp header line of the form:

```
[2026-03-23 01:17:27 GMT] [136.975] [-2.6/-27.3 dBFS] [24.7 dB] [-10.0 ppm]
```

followed by one or more content lines. A block ends immediately before the next timestamp header. The following fields within a block are inspected:

ICAO address

The first token on the line immediately after the timestamp, e.g. 407739 (Aircraft, Airborne) -> ...

AC info line

Registration is the first comma-separated field, e.g. AC info: G-TUMH, B38M, x521526028

ACARS Reg line

Registration may also appear in ACARS data with a leading dot that is stripped, e.g. Reg: .G-TUMH
Flight: BY0699

OUTPUT FILE

Matched blocks are written in raw text to a file in the current working directory. Blocks are separated by a single blank line. The filename is constructed as:

```
date4_LABEL.txt
```

where *LABEL* is determined as follows:

- If a **REG** was supplied, *LABEL* is the registration as given.
- If an **ICAO** was supplied and a registration is discovered anywhere in the matched blocks, *LABEL* is that registration.
- If an **ICAO** was supplied and no registration can be found, *LABEL* is the ICAO hex address.

EXAMPLES

Extract all records for registration G-TUMH from the 23 March log:

```
$ python3 extract_vdl2.py 0323 G-TUMH
Output: 0323_G-TUMH.txt
```

Extract records using ICAO hex address; registration auto-discovered:

```
$ python3 extract_vdl2.py 0323 407739
Output: 0323_G-TUMH.txt (REG discovered in log)
```

Extract by ICAO when no registration appears in the log:

```
$ python3 extract_vdl2.py 0323 A4E714
Output: 0323_N415DX.txt (REG found in AC info)
```

Extract records for an Irish-registered aircraft:

```
$ python3 extract_vdl2.py 0323 EI-IKR
Output: 0323_EI-IKR.txt
```

DIAGNOSTICS

[OK] N message block(s) written to: <filename>

Normal exit. N blocks matched and were written successfully.

[INFO] Registration discovered in log: <REG>

Printed after an ICAO search when a registration was found and used in the output filename.

[INFO] No records found for REG|ICAO = <value>

The identifier was not found in the log file. The script exits with status 0 and no output file is created.

[WARNING] Multiple files match ...

More than one .log file in the current directory matches the date string. The first match (alphabetical) is used.

[ERROR] No log file matching ...

No .log file containing the supplied date string was found in the current directory. Script exits with status 1.

EXIT STATUS

0 Success, or no matching records found.

1 A required log file could not be located.

FILES

pi21vdl2_YYYYMMDD.log

Typical input log file produced by dumpvdl2 running on the local receiver. The tool searches for any ***.log** file whose name contains the four-digit date argument.

date4_LABEL.txt

Output file written to the current working directory.

REQUIREMENTS

Python 3.9 or later. No third-party packages are required; only modules from the Python standard library (sys, os, re, glob) are used.

SEE ALSO

dumpvdl2(1) – VHF Data Link Mode 2 decoder. Project home: <https://github.com/szpajder/dumpvdl2>

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EXTRACT_VDL2(1)